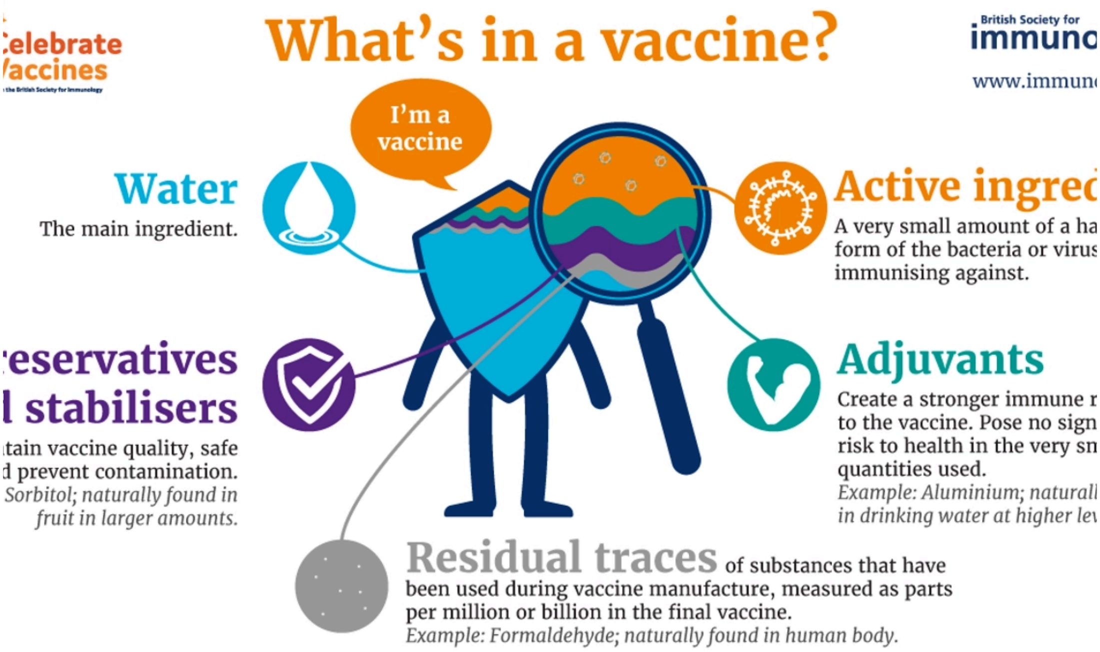
4.13 Vaccination

- What is a vaccine?
    - Something that **stimulates the immune system**, without causing serious harm or side effects
    - The aim is to **protect the individual against serious effects of disease (illness)** caused by a particular pathogen, **not to protect against actual infection by the pathogen.**
        - However, vaccinated individuals are unlikely to fall ill due to a pathogen, and hence are **unlikely to transmit the infection to others.**
    - Since the introduction of viruses into the population, **average life expectancies have increased** significantly globally.
- Describe how T cell mediated vaccine protection works.
    - The **innate immune response** leads to **pathogens being phagocytosed** and **antigens being presented** on the surface of macrophages/dendritic cells (APCs)
    - The **APCs present antigens to CD4 T cells** which leads to the **activation of these cells**
    - This leads to the **production of some CD8 killer and CD8 killer memory cells**, as well as some **B effector cells and B memory cells**, which overall **protect the patient against future exposure to the pathogen**, by means of **immune memory**
- What is Ro and what implications does it have on the spread of an infection?
    - **Ro = basic reproduction number** (i.e how many people can be infected by a single infected person)
    - **Ro < 1** means the **infection will likely die out** over time
    - **Ro > 1** means the infection is **likely to spread** across the population fast

▼ How does herd immunity work?

- **Person A** becomes infected by a disease, but **person B has been vaccinated against it**
- **Person A** comes into contact with **Person B**, but **Person B is immune to it**, and hence does not fall ill and cannot pass on the infection
- **Person B** comes into contact with **Person C**, but because Person B has been vaccinated, it means that Person C becomes indirectly protected from the virus, without even getting a vaccine
- Herd immunity describes how **one person becoming immune to a virus** can have a **large impact on the wider population.**

▼ Image showing the components of a vaccine.



▼ Why are viruses more difficult to grow than bacteria?

- Because **viruses are obligate parasites**, they require external machinery in order to be able to **grow and proliferate**
  - When **growing a virus**, you must grow the **pathogen with the cellular material**
- Bacteria however can be grown easily on an agar plate.

- What is an inactivated toxoid vaccine and describe the mechanism through which it provides immunity.
  - An **inactivated toxoid vaccine** is a vaccine which contains an **inactivated form of a toxin (toxoid)** secreted by a bacterial cell
  - **Antibodies** are produced which **recognise the shape of the toxin**, and hence they **bind onto the toxin**, **preventing it from binding onto the nerve** and **having negative effects.**
  - These antibodies help protect the individual in the case of subsequent exposure to the pathogen
  - An example is the **tetanus toxoid vaccine**, which **protects against tetanus**
- Describe an advantage and disadvantage of inactivated toxoid vaccines.
  - **Advantage** - They are cheap and safe
  - **Disadvantage** - only a small family of bacteria produce toxins, so these vaccines are limited
- What is a recombinant protein vaccine and describe the mechanism through which it provide immunity.
  - It involves the injection of a **viral spike protein** into the patient, in order to **induce an antibody response** against these proteins and **provide immunity**
    - This **viral spike protein is produced by inserting the relevant viral gene into a yeast cell** (viruses need hosts to proliferate)
  - It rests on the same principle as inactivated toxoid vaccines, but instead of toxoids, viral proteins are produced.
- Describe an advantage and disadvantage of recombinant protein vaccines.
  - **Advantage -** Pure and safe
  - **Disadvantages**
    - **Not all the proteins** made by viruses **are immunogenic** (can induce antibody responses).
    - The **proteins produced by the virus can also be a different shape** to the recombinant ones produced. (i.e RSV vaccine). This is because we are inserting genes into hosts and allowing the virus to grow here.
    - They are also relatively **expensive**

- Recall an advantage of using inactivated toxoid vaccines over recombinant protein vaccines.
    - They are cheaper
- Recall a disadvantage of using inactivated toxoid vaccines over recombinant protein vaccines.
    - Toxoid vaccines only work against a small family of bacterial cells.
- Why do we need an alternative method of bacterial immunity to viral immunity?
    - Viruses have spike proteins that can be easily used to induce antibody production through the **activation of B cells by CD4 T cells**
    - Most **bacterial cells have capsules** (polysaccharides, sugars) which are T independent antigens (B cells respond on their own). However, B cells are not very good at responding to these sugar antigens.
    - This means alternative methods are needed to provide bacterial immunity.
- What is this alternative method?
    - Conjugate vaccines
- What is an additional function of B cells, besides secreting antibodies?
    - B cells function as **professional antigen-presenting cells (APCs)** for CD4+ T cells
- What is a conjugate vaccine and describe the mechanism through which it provide immunity.
    - The **polysaccharide (sugar)** from the bacterial surface is **conjugated with a carrier protein**
        - This conjugate vaccine (sugar + protein) is injected into the patient
    - The **B cell recognises the polysaccharide** (the B cell hence has a B cell receptor that produces antibodies against the polysaccharide)
    - The **carrier protein** is taken into the B cell, and it **presents the antigens on its surface for CD4 T cells**. **Dendritic cells also do this.**
    - Tfh CD4 cells recognise the carrier protein on the B cell and release cytokines which allow the B cell to use its B cell receptor to produce the polysaccharide antibodies.
    - The **CD4 cells have helped to induce a larger antibody response within the B cells.**

- What is a dead pathogen vaccine?
    - A vaccine which involves the insertion of an entire dead pathogen, instead of just an antigen
    - The mechanism is still the same, in that antibody responses are still induced.
- What is a live attenuated vaccine?
    - A **weakened version of a live pathogen** used to induce a B cell response (i.e antibody production)
- What is an adjuvant and through which mechanism does it provide immunity?
    - It is a substance that is used to activate dendritic cells which increase the rate at which they activate T cells, and hence increase the rate at which B cells and antibodies are produced
    - It basically works by enhancing the innate response, in order to enhance to adaptive response.
    - Examples include **AS03** in **chickenpox** and **malaria vaccines** and **MF59** in **sequirius influenza vaccine.**
- Why are vaccines ineffective against viruses which show greater antigenic variability?
    - B and T cells (used to produce antibodies) recognise very small regions (~10 amino acids) on antigens
    - This means that it doesn’t take much (minor mutation) for a pathogen to evade the immune response produced against it
    - This can be seen as a **downside of** an (otherwise great) **adaptive immune system**
- Describe the phases involved in the development of vaccines.
    - **Preclinical** - Animals
    - **Phase 1** - Assessing safety
    - **Phase 2** - Assessing immunogenicity of the vaccine
    - **Phase 3** - Assessing effectiveness
    - **Phase 4** - Assessing for rare side effects - much larger population

▼ Which vaccines are given to children in the UK when they are eight weeks old?

- **DTaP/IPV/Hib/Hep B** I
- **Meningitis B** I
- **Rotavirus** I

▼ Which vaccines are given to children in the UK when they are twelve weeks old?

- **Pneumococcal conjugate vaccine** (PCV) I
- **DTaP/IPV/Hib/Hep B** II
- **Rotavirus** II

▼ Which vaccines are given to children in the UK when they are sixteen weeks old?

- **DTaP/IPV/Hib/Hep B** III
- **Meningitis B** II

▼ Which vaccines are given to children in the UK when they are one year old?

- **Hib booster/Meningitis C**
- **Measles, mumps and rubella (MMR)** I
- **Pneumococcal conjugate vaccine (PCV)** booster
- **Meningitis B** booster

▼ Which vaccines are given to children in the UK when they are two years old up to Year 6?

- **Nasal flu vaccine** (protects against seasonal flu)

▼ Which vaccines are given to children in the UK when they are 3 years and 4 months old?

- MMR booster
- Pre-school booster (4-in-1 vaccine against diphtheria, tetanus, pertussis (whooping cough) and polio)

▼ Which vaccines are given to children in the UK when they are teenagers?

- HPV vaccine
- Teenage booster (3-in-1 vaccine against tetanus, diptheria and polio)
- Meningitis ACWY (four types of meningitis)

- Which vaccines are given to older adults and risk groups?
    - Inactivated flu vaccine
    - Shingles vaccine
    - Pneumococcal polysaccharide vaccine (PPV)
- Which vaccines are given to pregnant women?
    - Inactivated flu vaccine
    - Pertussis (whooping cough)
- What does the **DTaP/IPV/Hib/Hep B (6-in-1)** vaccine protect against?
    - **Diphtheria, polio, tetanus, pertussis (whooping cough), Haemophilus influenzae Type B (Hib)** and **hepatitis B**
- What type of vaccine is Corynebacterium diphtheriae administered as?
    - Toxin based
- What type of vaccine is H. influenzae type B administered as?
    - Conjugate polysaccharide